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1.Introduction:

1.1 What is Sea-ice?

Sea-ice is the frozen ocean water which forms, grows, and melts in the ocean. In contrast, icebergs, glaciers, ice sheets, and ice shelves all originate on land. Sea-ice occurs in both the Arctic and Antarctic. In the Northern Hemisphere, it can currently exist as far south as Bohai Bay, China (approximately 38 degrees north latitude), which is actually about 700 kilometres (435 miles) closer to the Equator than it is to the North Pole. In the Southern Hemisphere, sea-ice only develops around Antarctica, occurring as far north as 55 degrees south latitude.

Sea-ice grows during the winter months and melts during the summer months, but some sea-ice remains all year in certain regions. About 15 percent of the world's oceans are covered by sea-ice during part of the year.

1.2 Formation and Growth of sea ice:

As the ocean water begins to freeze, small needle-like ice crystals called "frazil" form. These crystals are typically 3 to 4 millimetres (0.12 to 0.16 inches) in diameter. Because salt doesn't freeze, the crystals expel salt into the water, and frazil crystals consist of nearly pure fresh water. Sheets of Sea-ice form when frazil crystals float to the surface, accumulate and bond together.

Once Sea-ice forms into sheet ice, it continues to grow through the winter. When temperatures increase in spring and summer, the first-year ice begins to melt. If the ice does not grow thick enough over the winter, it will completely melt during the summer. If the ice grows enough during the winter, it thins

during the summer but does not completely melt. In this case, it remains until the following winter, when it grows and thickens and is classified as *multiyear ice*.

1.3 Characteristics of Sea ice:

Sea-ice is classified by stages of development that relate to thickness and age.

- New ice: Refers to ice less than 10 centimetres (3.9 inches) thick.
- Young ice: Ice that is 10 to 30 centimetres (3.9 to 11.8 inches) thick.
Young ice is sometimes split into two subcategories, based on colour:
 - 1) *grey ice* (10 to 15 centimetres, or 3.9 to 5.9 inches thick)
 - 2) *Grey-white ice* (15 to 30 centimetres, or 5.9 to 11.8 inches thick).
- *First-year ice*: thicker than 30 centimetres (11.8 inches), but has not survived a summer melt season.
- *Multi-layer ice*: Ice which survived a summer melt season and is much thicker than younger ice, typically ranging from 2 to 4 meters (78.7 to 157.5 inches) thick.

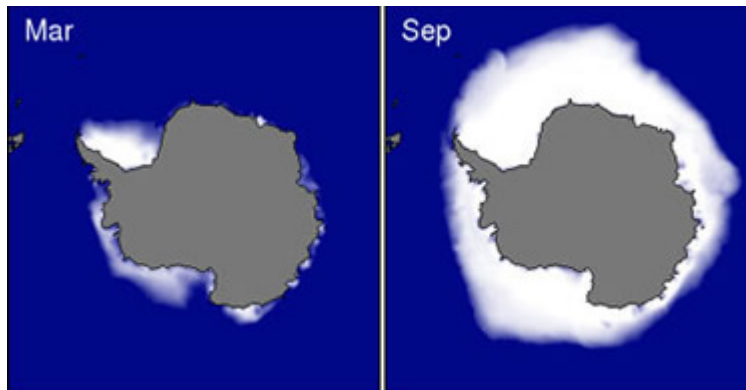


Figure 1 Antarctica sea ice

1.4 Why studies on sea-ice variability are important?

- There are chain of events occurs which starts a cycle of warming and melting of Sea-ice. This cycle is temporarily halted when the dark days of the polar winter return, but it starts again in the following spring. Even a small increase in temperature can lead to greater warming over time, making the polar regions the most sensitive areas to climate change on the Earth.
- Cold, dense, polar water sinks and moves along the ocean bottom toward the equator, while warm water from mid-depth to the surface travels from the equator toward the poles. Changes in the amount of Sea-ice can disrupt normal ocean circulation, thereby leading to changes in global climate.
- Too much or too little Sea-ice can be a problem for wildlife and people who hunt and travel in polar regions. In the Arctic, Sea-ice can be an obstacle to normal shipping routes through the Northern Sea route and Northwest Passage.
- Sea-ice is considered as one of the essential climate variable which modulates the albedo, reduces the ocean-atmospheric heat and gas

exchange, and plays a major role in ocean-atmospheric circulation and ecosystem. Sea-ice is a key component of Antarctic cryosphere which has a strong seasonal cycle with the mean extent varying from a minimum of $\sim 3.20 \times 10^6 \text{ km}^2$ in February to a maximum of $\sim 19.07 \times 10^6 \text{ km}^2$ in September. The influence of climate change through global warming on the Antarctic Sea-ice has a contrasting behaviour compared to that on the Arctic Sea-ice. The results based on 36 years of time series analysis of passive microwave satellite data (1979-2015) suggested significant expansion of Antarctic sea-ice extent at average rate of $24528 \pm 4306 \text{ km}^2/\text{yr}$ ($r = 0.69$, $p = 2.14 \times 10^{-6}$)

1.5 What is Polynya?

Polynyas are areas of persistent open water where we would expect to find sea ice. For the most part, they tend to be roughly oval or circular in shape, but they can be irregularly shaped, too.

The water remains open because of processes that prevent sea ice from forming or that quickly move sea ice out of the region. There are two types of polynyas described below, differentiated by the mechanism of ice removal.



Figure 2 Polynya

1.5.1 Sensible-Heat (Open-Ocean) Polynyas

A sensible-heat polynya forms when water that is above freezing upwells, or moves from the lower depths of the ocean to the surface. Heat transfers from the warmer water to the ice, melting it, and preventing new ice from forming.

The topography of the ocean bottom or overturning of water causes the warm water to rise to the surface. In this type of polynya, sensible heat from the ocean provides the source of heat needed to melt the ice.

Sensible-heat polynyas usually form in mid-ocean areas, far from coasts or other barriers.

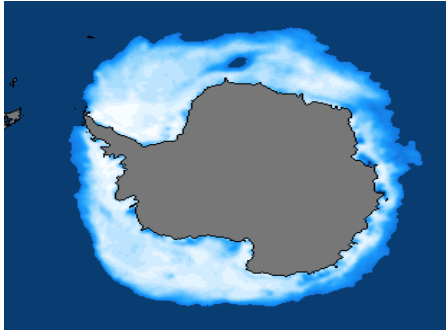


Figure 3 Sensible heat polynyas

1.5.2 Latent-Heat (Coastal) Polynyas

Latent-heat transfer occurs when matter changes state; latent heat is absorbed when ice melts, and it is released into the surroundings when liquid water freezes.

A latent-heat polynya is characterized by ocean water at the freezing point. It forms as a result of winds blowing in a persistent direction that push the ice away from a barrier, such as the coast, fast ice, a grounded iceberg or an ice shelf. As new ice grows within polynyas, wind blows it to the leeward side, while the windward side remains ice-free. Latent heat is released as water freezes and also as water evaporates into the air above the open water.

Some sensible-heat exchange also occurs within latent-heat polynyas because the water in the polynya is generally warmer than the air above it even though the water is at freezing temperature.

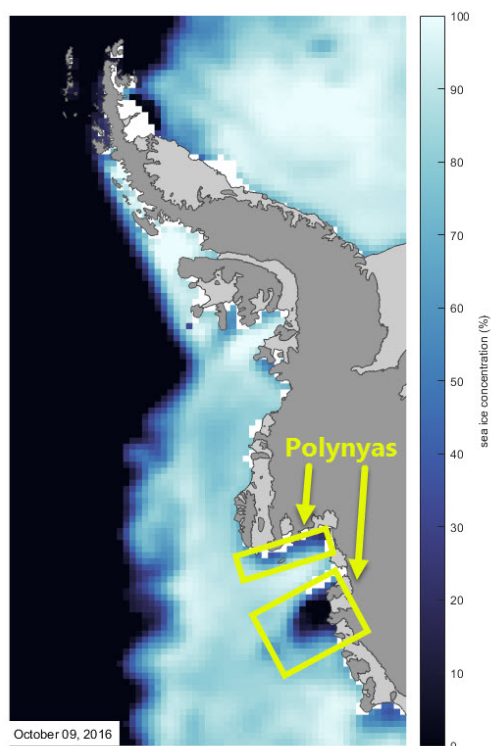


Figure 4 Latent heat polynyas

2. Objective

Develop processing methodology to study Antarctica sea ice variability.

3. Tasks and methods:

3.1 Task 1:

To identify polynya area formed in 2017 near Maud Rise and determining time period for polynya existence by using microwave remote sensing data and calculate polynya area for all dates.

3.1.1 Dataset: Sea Ice Concentration

Sea Ice Index images and alphanumeric data are based on gridded concentration data in the NSIDC Polar Stereographic Projection using the Hughes ellipsoid datum and EPSG 3411, for the Arctic, and EPSG 3412, for the Antarctic.

The Sea Ice time series is split into two sections: near-real-time and final.

The near-real-time section is comprised of the most recent data available and is created from the Near-Real-Time DMSP SSMIS Daily Polar Gridded Sea Ice Concentrations (NRTSI product) that is produced at and distributed from NSIDC.

The final section comprises data from the beginning of the record up to the most recent processing of the Sea Ice Concentrations from Nimbus-7 SMMR and

DMSP SSM/I-SSMIS Passive Microwave Data (GSFC product) that is produced at the Goddard Space Flight Center (GSFC).

Platform and Instrument	Time Period
Nimbus-7 SMMR	26 October 1978 - 20 August 1987
DMSP-F8 SSM/I	21 August 1987 - 18 December 1991
DMSP-F11 SSM/I	19 December 1991 - 29 September 1995
DMSP-F13 SSM/I	30 September 1995 - 31 December 2007
DMSP-F17 SSMIS	01 January 2008 - 31 March 2016
DMSP-F18 SSMIS	01 April 2016 - present

- Processing steps:

I) Obtain gridded input data

Processing begins with ice concentration data in the form of arrays, or gridded fields.

Daily data from different satellites mentioned above was used as input datasets for generating near-real-time as well as final data.

Monthly geotiff data of respective type was constructed using respective product of daily concentration data.

II) Derive concentration fields

Sea Ice Index processing creates the daily concentration fields from the input product mentioned above.

A threshold value of 15 percent is used as a cut off for the lowest ice concentration because data values less than 15 percent from passive microwave instruments are too uncertain to use. Therefore, values less than 15 percent are set to be 0 percent ice and the rest of the data ranges from 15 percent to 100 percent concentration.

Monthly averages are computed from the daily product data for that cell only if there are at least 20 days of data for that cell. Otherwise, the cell is labelled as missing.

III) Scale data.

Before creating the concentration GeoTIFFs, the data are scaled by 10 so that colour table entries can be created for each value because colour tables must be indexed by an integer value.

IV) Create GeoTIFFs.

GeoTIFF creation is accomplished using Python bindings to the Geospatial Data Abstraction Library(GDAL). For the concentration GeoTIFFs, these are created using unsigned 16 bit values with associated geographic metadata and color table.

- Source data resolution:

Source data grid cells are nominally 25 km by 25 km, but this is not the same as the resolution of the satellite instrument sensor channels used to create source data ice concentration products. That resolution, or field of view (FOV), is an

ellipsoid with a long axis that ranges from about 28 to 69 km, depending on channel frequency and instrument series.

Instrument	Frequency (GHz)	FOV (km)
SMMR	18.0	55 x 41 ⁽¹⁾
	37.0	27 x 18 ⁽¹⁾
SSM/I	19.35	69 x 43 ⁽²⁾
	37.0	37 x 28 ⁽²⁾
SSMIS	19.35	74 x 45 ^(3, 4)
	37.0	45 x 28 ^(3, 4)

- As a result of the task from daily data analysis, polynya was found for following dates in different years

Year	From	To
1974	20 th July	13 th December
1975	Data not available	24 th November
1976	30 th July	1 st November
1994	18 th September	20 th September
2017	13 th September	2 nd December

- This dataset was re-projected from 'south-polar stereographic (EPSG: 3412)' to 'Geographic WGS-84(EPSG: 4326)' using GDAL software.

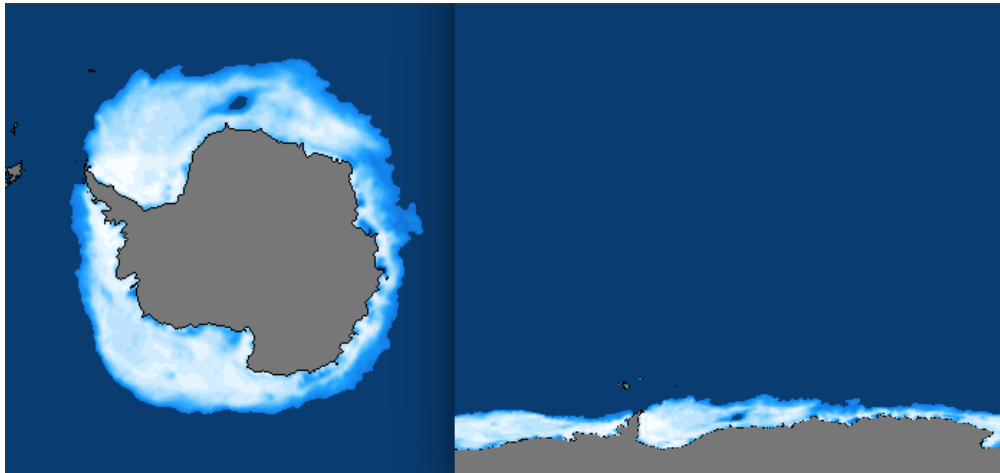


Figure 5 Reprojection

South polar stereographic (EPSG: 3412) GCS WGS-84 (EPSG: 4326)

- Mapping of selected dates for all years with same colour coding was done using QGIS software.
- To observe the nature of polynya during its existing time period, subset of maximum polynya area extent was taken and area was calculated using python programming.

Area = number of pixels having SIC $\leq$ 15% x resolution

- It was observed that polynya area is gradually increasing for all years.

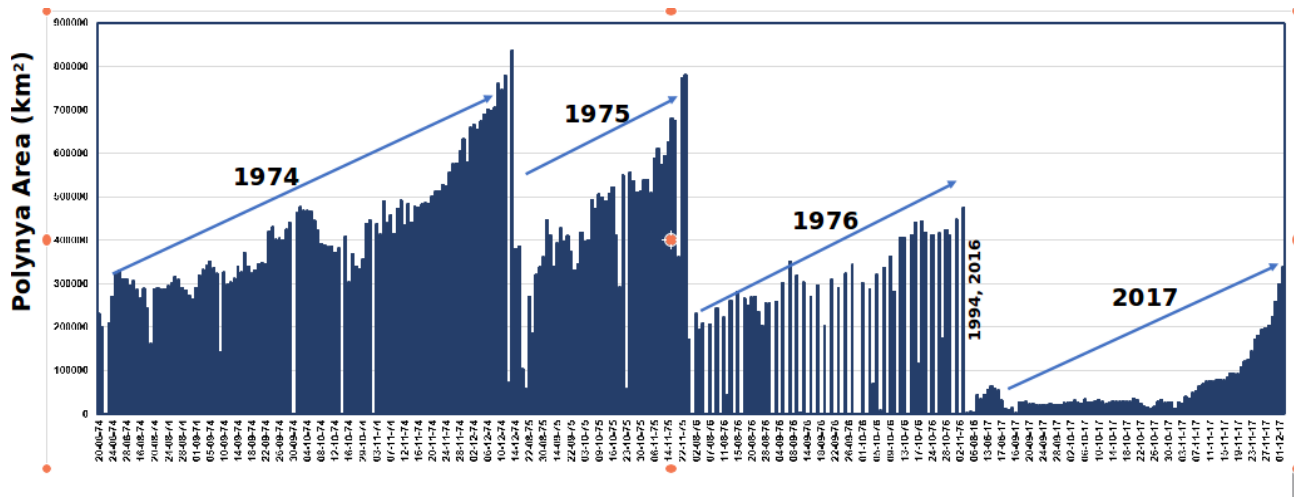


Figure 6 Area Calculation

3.2 Task 2:

Download daily re-analysis data from ECMWF ERA-interim data of sea ice cover, mean sea level pressure and wind direction to compare polynya area with microwave data and to understand wind direction during the time of polynya as well as few months prior to polynya formation.

3.2.1 Dataset: Sea Ice Cover, Wind (u,v), Mean sea level pressure

ERA-Interim is a reanalysis of the global atmosphere covering the data since 1979.

The ERA-Interim data assimilation and forecast suite produces:

- Four analyses per day, at 00, 06, 12 and 18 UTC
- Two 10-day forecasts per day, initialized from analyses at 00 and 12 UTC

The analysis produced at 00 UTC on a given day involves observations taken between 15 UTC on the previous day and 03 UTC on the present day; the analysis at 12 UTC involves observations between 03 UTC and 15 UTC.

Fields from the atmospheric model are archived either at the full T255 spectral resolution or on the corresponding N128 reduced Gaussian grid, depending on their basic representation in the model. Fields from the coupled ocean-wave model are saved on its reduced 1.0degx1.0deg latitude/longitude grid.

To subset and merge the netcdf data downloaded from ECMWF, ‘climate data operator’ (CDO) was used and to make a spatial plot as shown below, ‘Grid Analysis and Display System (GrADS)’ was used.

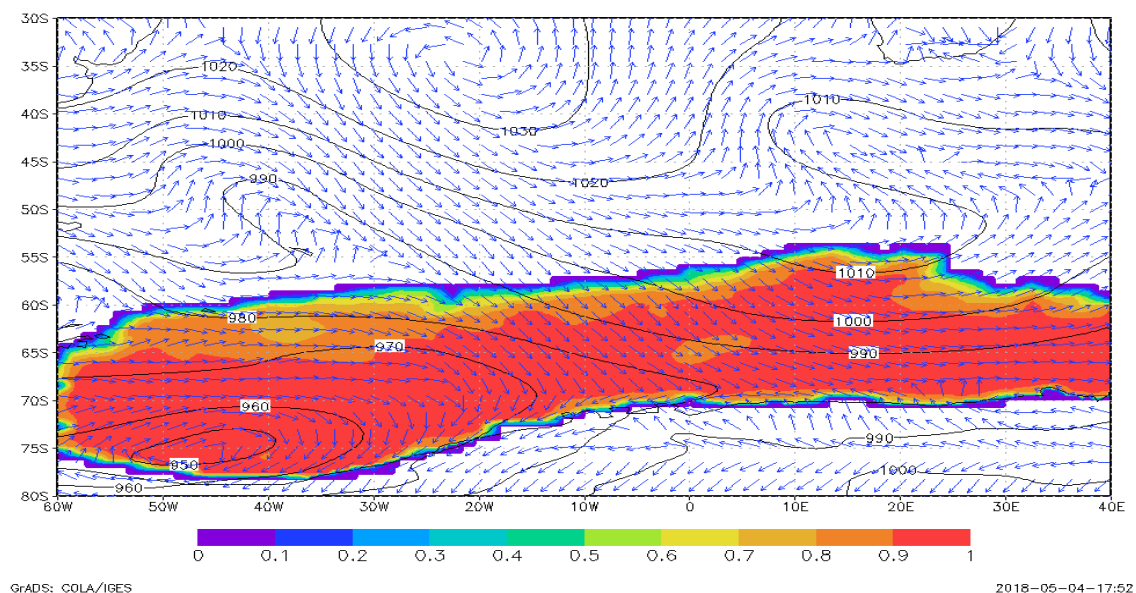


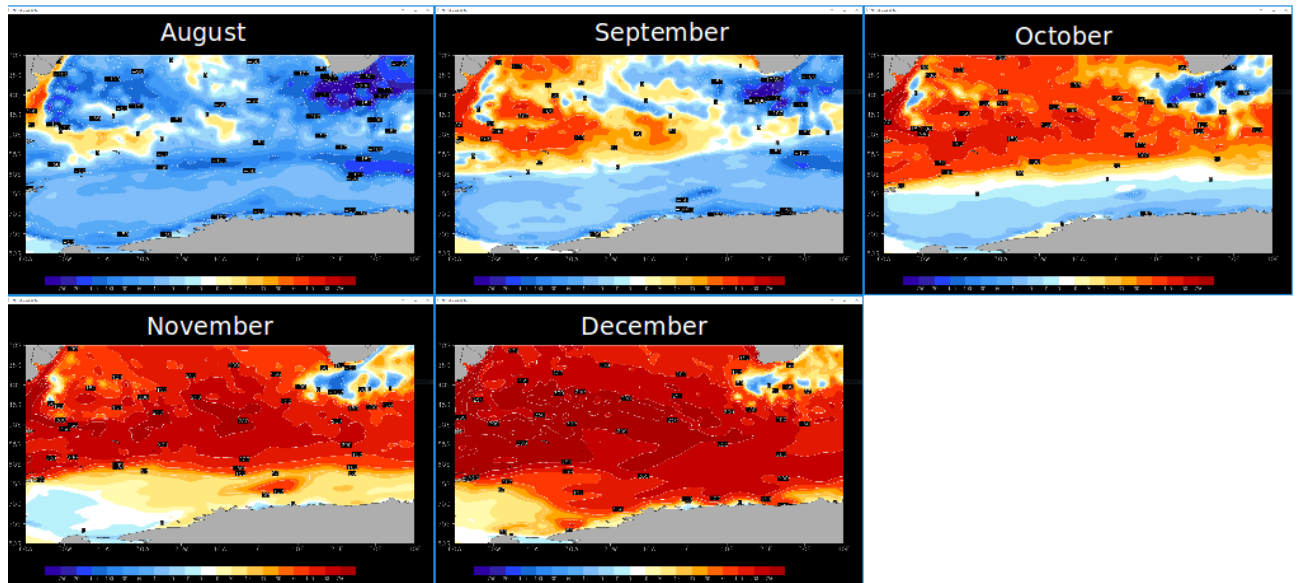
Figure 7 Atmospheric pressure, wind and sea-ice on 18-September-2017

3.3 Task 3:

Download daily and monthly re-analysis data of following parameters from ECMWF to calculate monthly anomaly using CDO.

Resolution is 0.25 deg x 0.25deg for all parameters.

Parameter name	Unit
Sea surface temperature	K converted to degree Celsius
2m temperature	K converted to degree Celsius
Mean sea level pressure	Pa converted to hpa
10m wind components(u,v)	m/s
Vertical integral of eastward/northward water vapour flux	Kg/ms
Vertical integral of water vapour	Kg/m ²
Surface sensible heat-flux	J/m ² converted to degree Celsius watt/m ²
Surface latent heat-flux	J/m ² converted to degree Celsius watt/m ²
Surface net thermal radiation	J/m ² converted to degree Celsius watt/m ²
Surface net solar radiation	J/m ² converted to degree Celsius watt/m ²



➤ Unit conversion formulas:

$$\text{Temperature (Celsius)} = \text{Temperature (K)} - 273.15$$

$$1 \text{ hPa} = 100 \text{ Pa}$$

$$1 \text{ Watt/m}^2 = 1 / (6 \times 3600)$$

➤ Anomaly calculation was done using CDO software using following formula:

➤ September2017 anomaly = Absolute values of september2017 - Climatology for September

Where,

Climatology for September = mean (September1979..... September2015)

3.4 Task 4:

To identify values of the year 2017 which are lower than the lowest value of years 1979 to 2016 as well as values which are higher than the highest value of years 1979 to 2016, for all parameters mentioned in previous task.

This task was done month wise for 2017 data using CDO software.

- Step 1: merge the data from year 1979 to 2016.
- Step 2 : take minimum and maximum over all years for each month(September, October, November, December)

Lowest for September (a) = minimum (sept-1979....sept-2016)

Highest for September (b) = maximum (sept-1979....sept-2016)

- Step 3: Select the values from 2017 data which are lower than 'a' and higher than 'b'. Output will be the desired values.

In output shown below, grey coloured lines are highlighting extreme low and extreme high values.

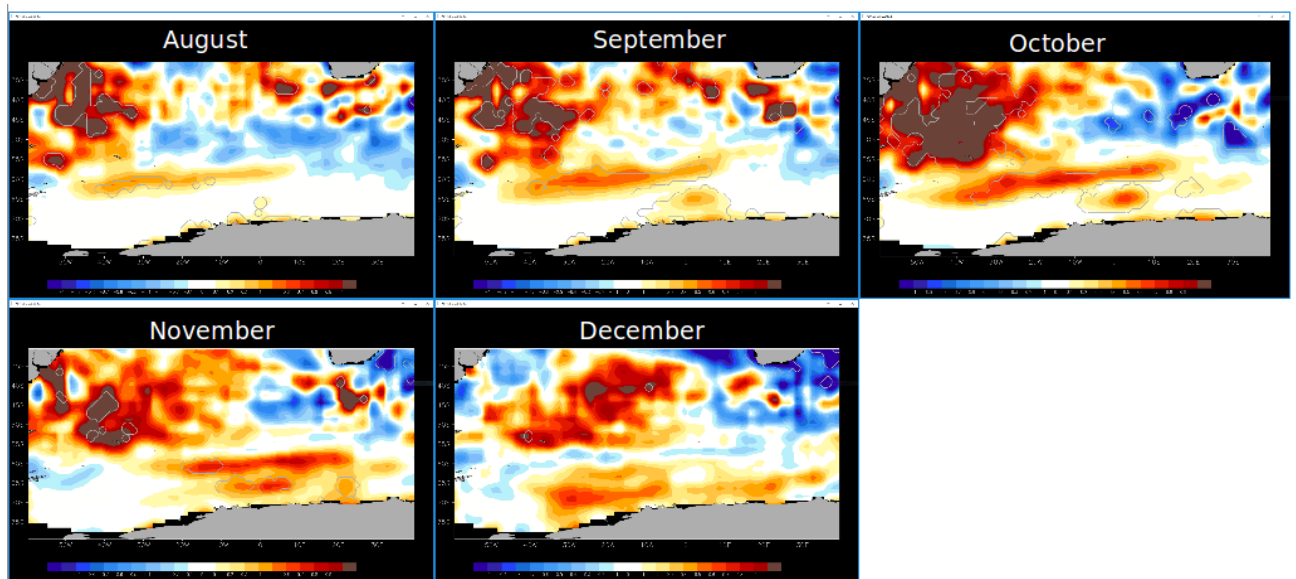
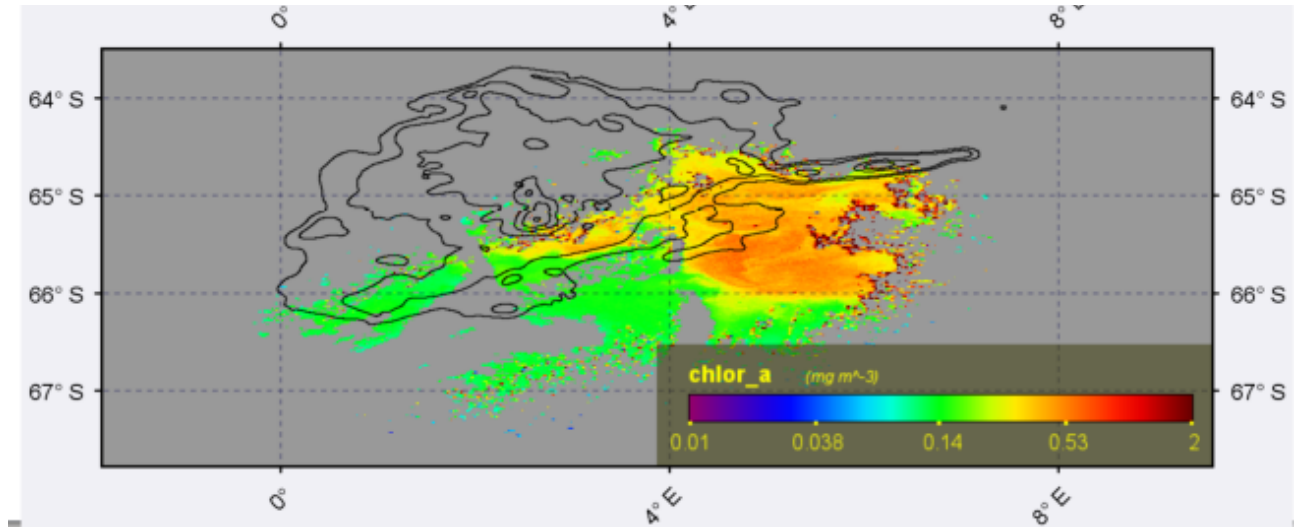


Figure 9 Extreme low and extreme high values

3.5 Task 5:

This task is to download following parameters of VIIRS and MODIS satellites for the months of October, November and December to analyse presence of chlorophyll in polynya area.

- Monthly data for the month of September was not available because of cloud cover. For few days in September data is available for chlorophyll.



Parameter name	Unit
Chlorophyll concentration	Mg/m ³
Euphotic depth	m
Diffuse attenuation coefficient at 490 nm	m ⁻¹
Photosynthetically Available Radiation	einstein m ⁻² day ⁻¹
Net primary production	mgC m ⁻² day ⁻¹

3.5.1 Dataset: MODIS and VIIRS data (4km resolution)

We have used level-3 data of MODIS and VIIRS for analysis. Level 3 data are derived geophysical variables that have been aggregated/projected onto a well-defined spatial grid over a well-defined time period.

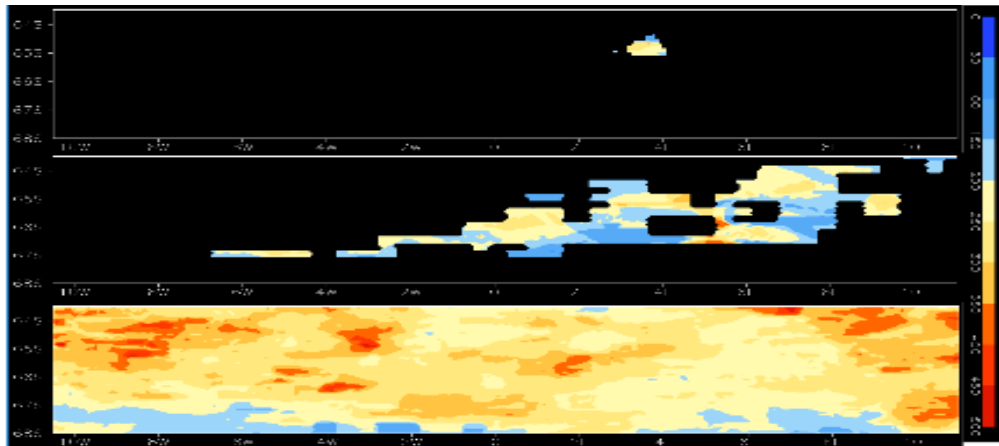


Figure 11 Aqua-MODIS Log (Chlorophyll-a) _mg/m3

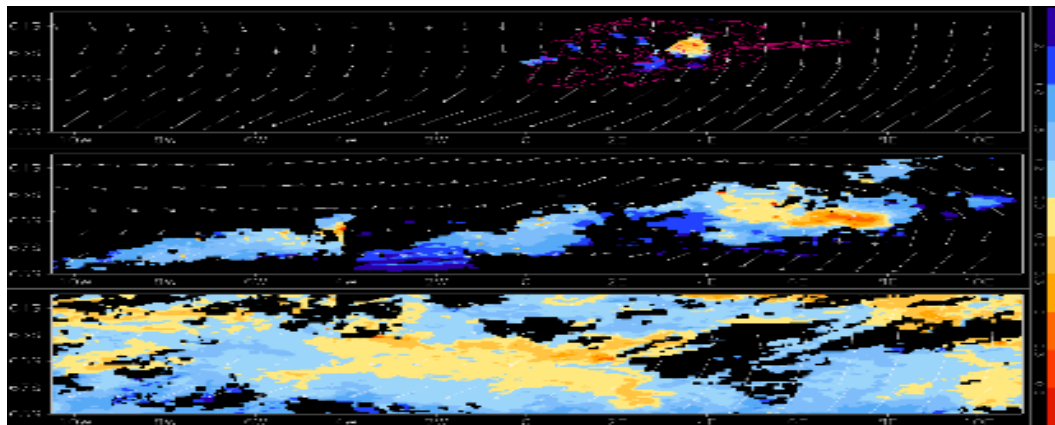


Figure 12 VGPM monthly net primary production (mg Carbon/m2/day)

3.6 Task 6:

To download and process sea ice drift data to overlay upon sea ice concentration microwave data.

3.6.1 Dataset: Sea ice drift (62.5 km resolution)

Following sensors and corresponding channels are used to generate merged sea ice drift data.

Sensor	Platform	Channels	Sampling [km]	Footprint [km]	Last active date
SSMIS	DMSP-F18	91 GHz, H+V pol.	12.5	14x16	Active
ASCAT	Metop-B	C band σ^0	12.5	(25-34)x(25-34)	Active
AMSR2	GCOM-W1	37 GHz and 18.7 GHz H+V pol.	10	12x7 and 14x22	Active
SSMIS	DMSP-F17	91 GHz, H+V pol.	12.5	14x16	fall 2016
ASCAT	Metop-A	C band σ^0	12.5	(25-34)x(25-34)	fall 2016
AMSR-E	EOS Aqua	37 GHz, H+V pol.	10	14x8	Oct 3 rd 2011
SSM/I	DMSP-F15	85 GHz, H+V pol.	12.5	14x16	Jan 19 th 2013

The 10 km gridded NH (SH) operational sea ice edge product is remapped to a 12.5 km resolution polar stereographic grid to use as sea ice mask.

Swath files are remapped and averaged on a 12.5 km polar stereographic grid. When several channels are present for a given instrument (e.g. two polarizations) they are kept in the same file. The average sensing time for each pixel is also recorded.

- Re-projection was required here because all other datasets are in ‘GCS-WGS84’ projection, to overlay this data on sea ice concentration it was required to re-project it from south polar stereographic projection to GCS-WGS84.
- To re-project it using GDAL it was required to georeference it. We have to georeference dx,dy(drift vectors) using lat,lat1,lon and lon1 subdatasets.
- Converted each subdataset to virtual raster and georeferenced (dx,dy) using GDAL.

- Georeferenced data was then re-projected and re-gridded to match the resolution of SIC data.

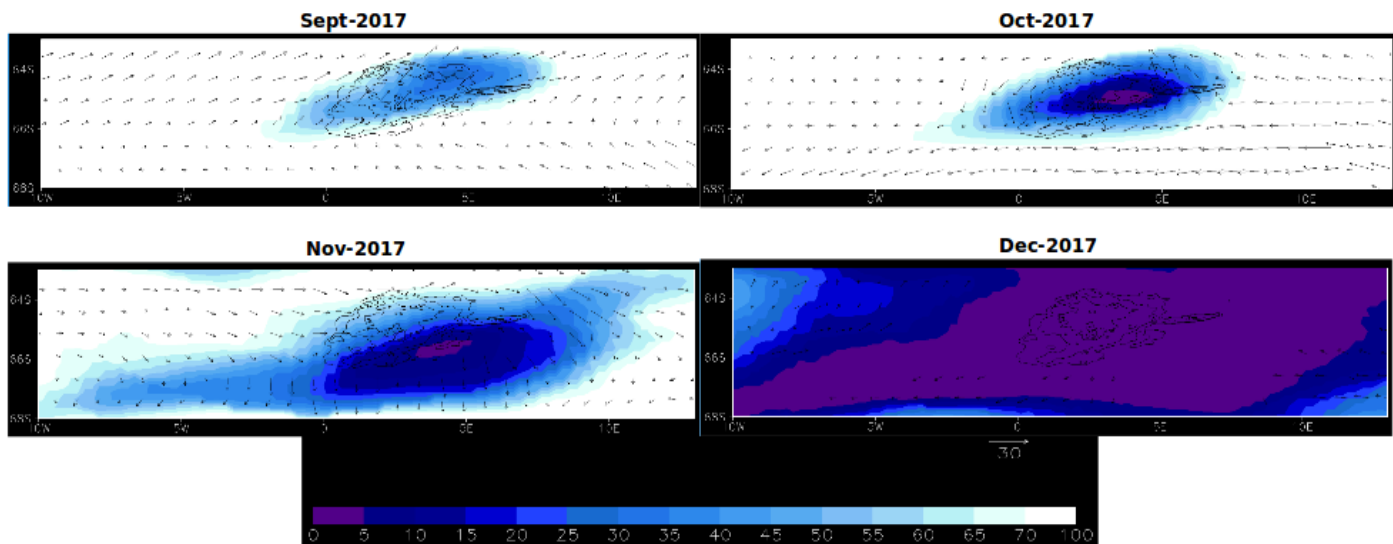


Figure 13 Sea-ice vector overlaid on SSMI/S sea-ice concentration

3.7 Task 7:

To download and process ‘Absolute dynamic topography’ data to understand ADT and current vector of the polynya area.

3.7.1 Dataset: Absolute dynamic topography

Absolute dynamic topography(ADT) is an along-track multi-mission altimeter product. It gives along-track sea surface heights above geoid. It is the sum of sea level anomaly (SLA) and mean dynamic topography (MDT).

It is a product of processed data from all altimeter missions: Jason-3, Sentinel-3A, HY-2A, Saral/AltiKa, Cryosat-2, OSTM/Jason-2, Jason-1, Topex/Poseidon, Envisat,GFO, ERS-1&2.

The Exponents and Orientations are distributed on a $1/25^\circ$ grid on a global coverage $0^\circ\text{E}/359.96^\circ\text{E}/65^\circ\text{S}/85^\circ\text{N}$.

In Delayed-Time, the maps are calculated every 4 days since 1994/04/01. They are updated every year. In Near-Real-Time, the maps are delivered every day, with 20 days of delay.

■ **Input parameters:**

I) The Altimetry gives access to the Sea Surface Height (SSH) above the reference ellipsoid.

$$\text{SSH} = \text{Orbit} - \text{Altimetry Range}$$

II) The Mean Sea Surface (MSS) is the temporal mean of the SSH over a period N. It is a mean surface above the ellipsoid and it includes the Geoid.

$$\text{MSSN} = \langle \text{SSH} \rangle N$$

III) Sea Level Anomaly (SLA) is deduced from the SSH using a Mean Sea Surface (MSS)

$$\text{SLAN} = \text{SSH} - \text{MSSN}$$

IV) Mean Dynamic Topography (MDT) is the temporal mean of the SSH above the Geoid over a period N.

$$\text{MDTN} = \text{MSSN} - \text{Geoid}$$

Hence the product used here is derived using:

$$\text{ADT} = \text{SLAN} + \text{MDTN} = \text{SSH} - \text{MSSN} + \text{MDTN}$$

- Task was to clip the polynya area for all months using shapefile of maximum polynya extent for each month to avoid erroneous data.
- Export the data in text format to plot it using ODV(Ocean data visualisation) software.
- Plot ADT and overlay geostrophic current data over it.

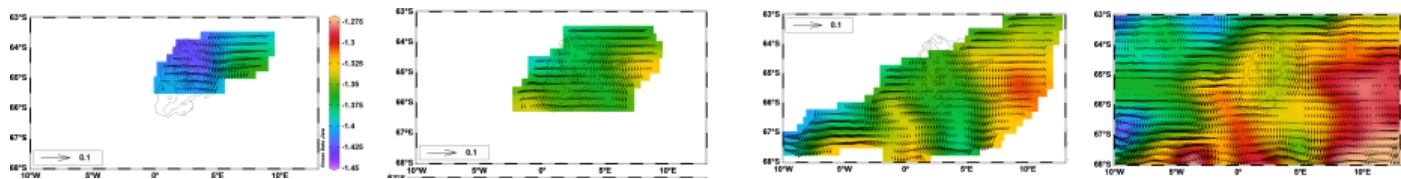


Figure 14 Monthly images of Absolute dynamic topography

3.8 Task 8:

- To download and process sea surface temperature data product of microwave+ infrared band and process it to compare the outputs with OISST.
- Calculate the anomaly and extreme low/high value for the dataset.
- Export minimum, maximum and mean values to excel and compare the graph with oisst data as well as ECMWf sst data.

3.8.1 Dataset: sea-surface temperature (mw+ ir):

This product is an optimally interpolated (OI) SST product which is created from the microwave (MW) and infrared (IR) SSTs. This is a global data with the resolution of 9km available from June-2002 to present.

This product is generated using bunch of microwave as well as infrared sensors as mentioned below.

MW: TMI, AMSR-E, AMSR-2, WindSat, GMI

IR: MODIS-Terra, MODIS-Aqua, VIIRS-NPP

SSTs from different satellites are blended together using the optimum interpolation (OI) scheme to generate the final product. The OI technique requires the specification of data and analysis error statistics.

A first-guess field, the previous day's OI SST, is employed to calculate data increments, which are all nearby data minus the first-guess field. The new SST estimate is formed by a weighted sum of increments, with the weights calculated by the OI method, added to the first guess data. Correlation scales of 3 days and 100 km are used in determining the weights used in here in the methodology for generating the final product.

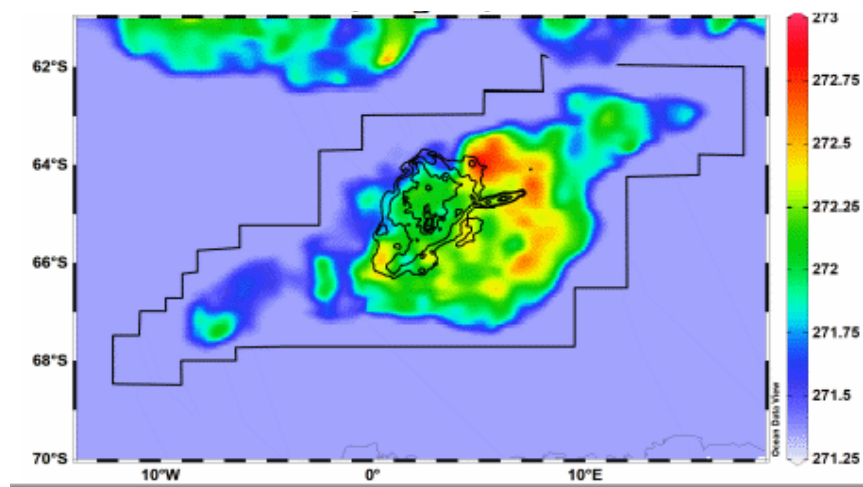


Figure 15 MW+IR SST on 3rd December 2017

3.9 Task 9:

Download and process wind stress dataset for the month of september, october, november and december to analyse wind stress and other components on and around polynya area.

3.9.1 Dataset: Windstress, Upwelling, Curl

NOAA CoastWatch distributes science quality wind stress data in zonal, meridional, modulus, and wind stress curl sets. This data begins with wind velocity measurements from the ASCAT instruments onboard all EUMETSAT's MetOp satellites. ASCAT is a microwave scatterometer designed to measure surface winds over the global ocean. NOAA CoastWatch processes wind velocity to wind stress and wind stress curl. (This is a 1 day composite.)

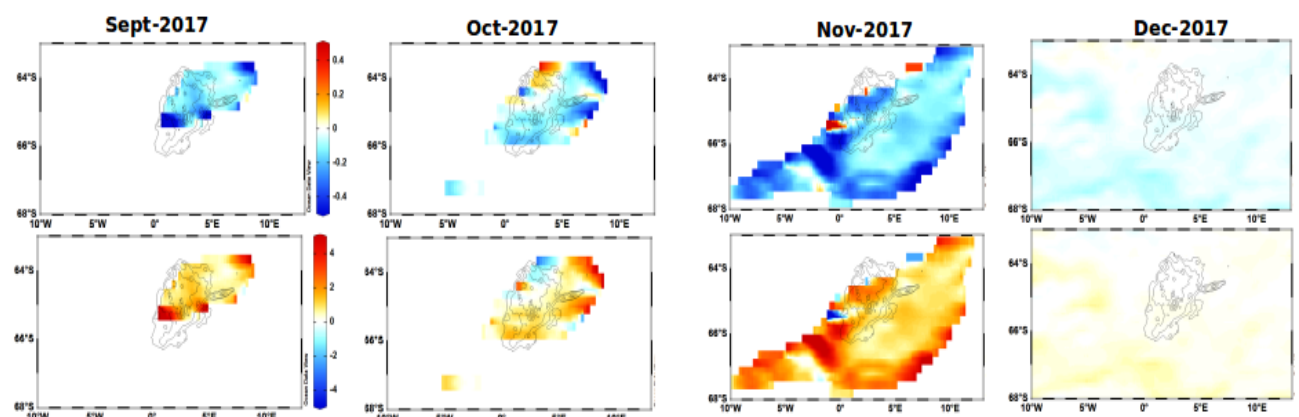


Figure 16 Upwelling and Windstress

3.10 Task 10:

To analyse ocean parameters such as ocean temperature, current and salinity on and around polynya area.

3.10.1 Dataset: Ocean temperature, Current and Salinity

Ocean Reanalysis System 4 (ORAS4) data gives global coverage with spatial resolution of 1 degree. It is generated from daily surface fluxes were derived from the operational ECMWF atmospheric analysis.

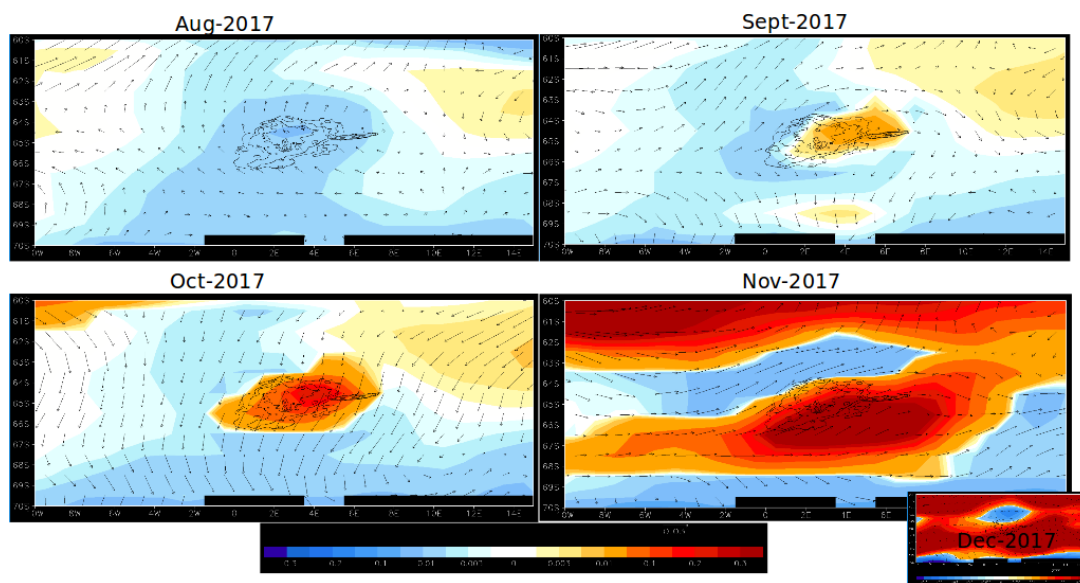


Figure 17 Subsurface ocean current (m s⁻¹) and temperature (deg C) anomaly below sea-ice

3.11 Task 11:

To create line graphs of all parameters to analyse similarity between their trends as well as observe trend in the polynya region.

Plot line graphs using 'singmaplot' and find cross correlation matrix of all parameters and datasets had been used.

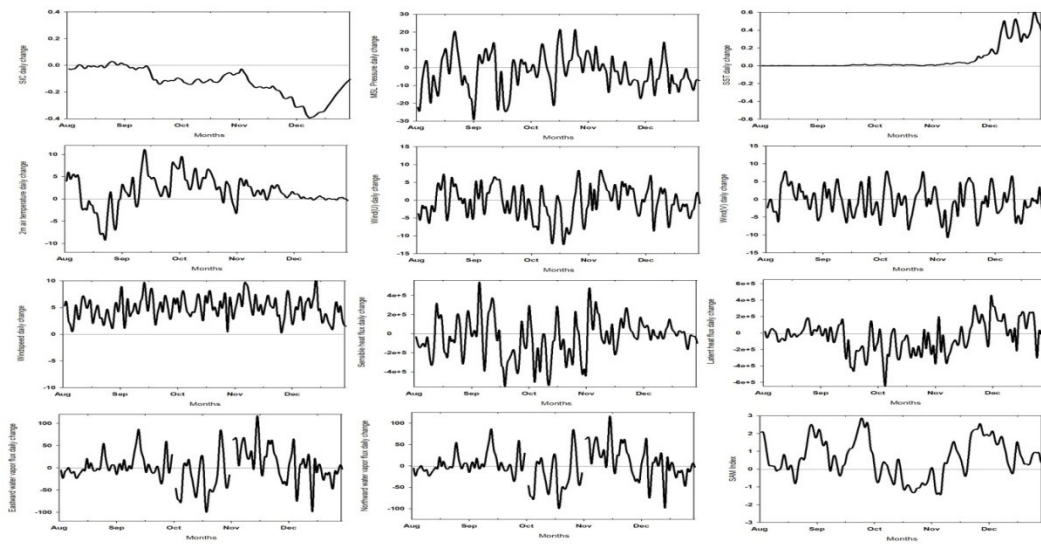


Figure 18 Graphs for analysis

4. Results and discussion:

The top of the Maud Rise seamount is located at 2.63°E , -65.23°S , which rises from the abyssal plain of ~ 5200 m to the shallowest depth of ~ 968 m.

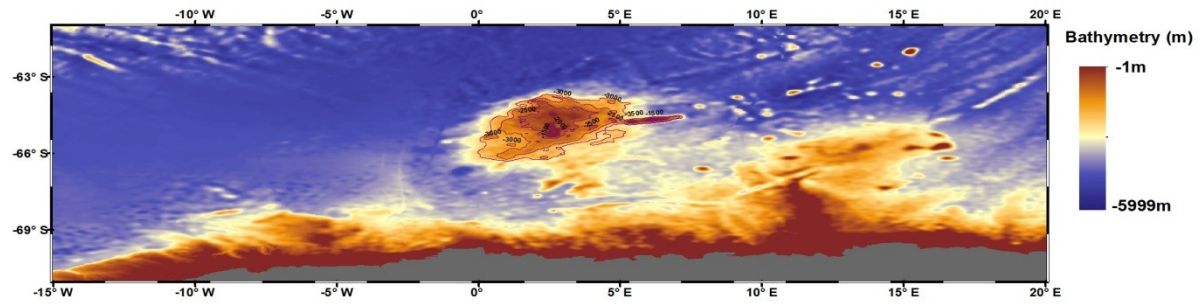


Figure 19: Bathymetry

Analysis of sea-ice climatology indicated low sea ice concentration prevails above the Maud Rise seamount, Eastern Weddell Sea, for the months August through December.

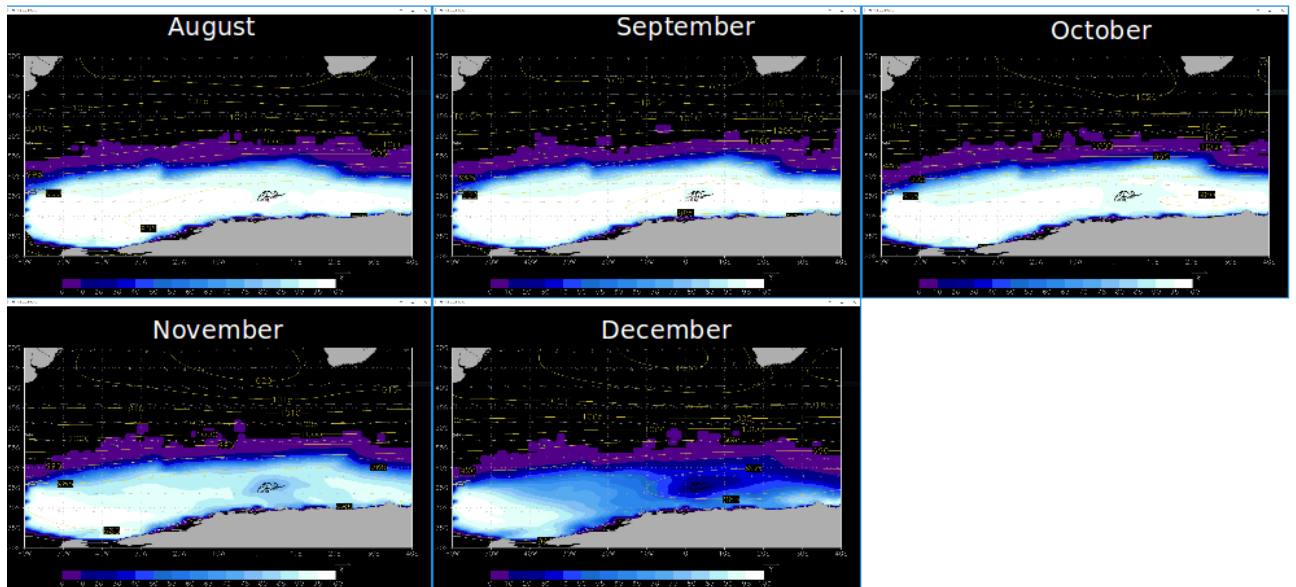


Figure 20: Sea ice Climatology

The launching of first passive microwave sensor, ESMR (Electrically Scanning Microwave Radiometer) onboard the Nimbus-5 satellite could observe the polynya on the flanks of Maud rise seamount during 1974-1976.

The polynya first observed during 20-July-1974 with an areal extent of 0.23 million km² which sustained up to 13-December-1974. On 18-August-1975, the polynya reappeared with an areal extent of 0.38 million km² and sustained up to 24-November-1975. On 30-July-1976, the polynya observed with an areal extent of 0.17 million km² which expanded (~4.1 million km²) and sustained till 01-November-1976. In each case of events during 1974-76, the polynya was formed initially on the eastern flanks of Maud rise seamount which enlarged dominantly across western flanks. The polynya was formed during 1974-1976, not through the melting of any pre-existing ice within ice-covered zone, but by preventing the growth of new ice within ice coverage. After 1976, the event was absent almost for next 18 years except persistent low SIC (70-90%) on the seamount compared to its periphery SIC (100%). On 19th September 1994, a small size polynya (1250 km²) was appeared for one day, formed to the north-east flanks of the seamount. After 22 years break, it appeared on 5th August 2016 for two days covering with an area of 4375 km².

In each case of 1994 and 2016, the polynya was formed through the melting of pre-existing ice within ice-covered zone. The polynya was appeared again on 10th June 2017 with areal extent of 43125 km², formed by preventing the growth of new ice. This event could sustain for nine days (up to 18th June 2017), and disappeared after the growth of new ice in austral winter. Finally, on 04-December-2017, the ice-edge having low SIC was broken at the northern-eastern boundary of polynya which merged with the open ocean due to the SST warming caused by net gain of heat flux by the sea surface. Its northern boundary was completely disappeared before 10-December-2017, due to southward intrusion of warm water into its vicinity. In each case, the polynya merged with the sea through the sea-ice deformation to the north.

5. Tools and codes:

5.1 To calculate anomaly:

To take daily mean:

```
for((i=10,j=17;i<=j;i++))
```

```
do
```

```
  cdo ydaymean "$i"_aug.nc "$i"_aug_mean.nc
```

```
  cdo ydaymean "$i"_sept.nc "$i"_sept_mean.nc
```

```
cdo ydaymean "$i"_oct.nc "$i"_oct_mean.nc
```

```
cdo ydaymean "$i"_nov.nc "$i"_nov_mean.nc
```

```
cdo ydaymean "$i"_dec.nc "$i"_dec_mean.nc
```

Done

Daily anomaly:

```
cdo -b 64 mergetime *.nc 79_15.nc
```

```
cdo ydaymean 79_15.nc climatology.nc
```

```
cdo -b 64 mergetime 17_*.nc 2017.nc
```

#note : delete extra time steps and keep 153 timesteps climatology and 2017

```
cdo delete,timestep=122 2017.nc del.nc
```

#note:delete 2017.nc and rename final output with 153 steps as 2017nc from del.nc

```
cdo sub 2017.nc climatology.nc anomaly.nc
```

Monthly anomaly:

```
cdo ymonmean 79_15.nc climatology.nc
```

```
cdo -b 64 sub 17.nc climatology.nc anomaly.nc
```

5.2To calculate extreme minimum and extreme maximum values(hash values) :

```
cdo runmax,38 oct_79_16_ci.nc oct_max_79_16_ci.nc
```

```
Cdo -b 64 -gt aug_17_sst.nc aug_max_79_16_sst.nc aug_17_sst_hashmax.nc
```

```
cdo shifttime,1days oct_17_ci_hashmax.nc max_17_ci.nc
```

```
cdo cat oct_17_ci_hashmin.nc max_17_ci.nc oct_ci_hash.nc
```

5.3 To reproject the raster:

```
gdalwarp -overwrite -s_srs EPSG:3412 -t_srs EPSG:4326 -te -180 -90 180 90  
-tr 0.25 0.25 S_201712_concentration_v3.0.tif sic_dec.tif
```

5.4 To Calculate polynya area:

```
import numpy as np  
  
import csv  
  
import re  
  
from osgeo import gdal  
  
import ogr  
  
from osgeo import gdal_array  
  
from gdalconst import *  
  
import os  
  
import osr  
  
import shutil  
  
import subprocess  
  
indir = '/home/vigna/Documents/polynyas/SIC/1972-  
77/p0lynya_74_75_76/1975'  
  
directory=os.fsencode(indir)  
  
e=1
```

```

z=0

for file in os.listdir(directory):

file1=os.fsdecode(file)

if '_subset.tif' in file1:

name=str(e)+'_subset.tif'

#print(name)

driver = gdal.GetDriverByName( 'GTiff')

file = gdal.Open(os.path.join(indir,name),GA_ReadOnly)

band = file.GetRasterBand(1)


lista = band.ReadAsArray()
for j in range(file.RasterXSize-1):
for i in range(file.RasterYSize-1):

values = lista[lista>=200]

```

```
za = len(values)

#if lista[i,j] <= 150:

#print(len(lista[i,j]))

#z+=1

#print(za)


e+=1

a=(za*25*25)

print(a)
```

5.5 Drift data processing:

```
#cdo splitmon drift_monthly.nc mon

for((i=10,j=12;i<=j;i++))
do

gdal_translate -of VRT NETCDF:"mon"$i".nc":lat "$i"_lat.vrt
gdal_translate -of VRT NETCDF:"mon"$i".nc":lon "$i"_lon.vrt
gdal_translate -of VRT NETCDF:"mon"$i".nc":lat1 "$i"_lat1.vrt
gdal_translate -of VRT NETCDF:"mon"$i".nc":lon1 "$i"_lon1.vrt
gdal_translate -of VRT NETCDF:"mon"$i".nc":dX "$i"_dx.vrt
gdal_translate -of VRT NETCDF:"mon"$i".nc":dY "$i"_dy.vrt


gdalwarp -geoloc -t_srs EPSG:3412 "$i"_dx.vrt "$i"_dx.tif
gdalwarp -geoloc -t_srs EPSG:3412 "$i"_dy.vrt "$i"_dy.tif


gdalwarp -overwrite -s_srs EPSG:3412 -t_srs EPSG:4326 -te -180 -90 180 90
-tr 0.25 0.25 "$i"_dx.tif "$i"_dx_reproject.tif

gdalwarp -overwrite -s_srs EPSG:3412 -t_srs EPSG:4326 -te -180 -90 180 90
-tr 0.25 0.25 "$i"_dy.tif "$i"_dy_reproject.tif


gdal_translate -of netcdf "$i"_dx_reproject.tif "$i"_dx.nc
```

```
gdal_translate -of netcdf "$i"_dy_reproject.tif "$i"_dy.nc
```

```
cdo merge "$i"_dx.nc "$i"_dy.nc "$i"_dx_dy.nc
```

Done

5.6 To plot in GRADs:

```
'sdfopen anomaly.nc'
```

```
'set grads off'
```

```
'set ylopts 1 3 0.14'
```

```
'set xlopts 1 3 0.14'
```

```
'set t 2'
```

```
'run ci_rgb.gs'
```

```
'q pos'
```

```
'set gxout contour'
```

```
'set cint 2'
```

```
'set ccolor 0'
```

```
'set clab forced'
```

```
'set clopts -1 -1 0.14'
```

```
'd msl_hpa'
```

```
'set ccolor 4'
```

```
'set arrscl 0.4 5'
```

'd skip(u10,8);skip(v10,8)'

'sdfopen sept_hash.nc'

'set dfile 2'

'set t 1'

'set gxout contour'

'set clab off'

'set ccolor 2'

'd ci'

'set gxout contour'

'set clab off'

'set ccolor 15'

'd msl'

'set t 2'

'set gxout contour'

'set clab off'

'set ccolor 15'

'd msl'

'set line 0'

'set shpopts 15'

'draw shp ATA_adm0.shp'

6. Learning

6.1 Work experience:

I had learned different GIS softwares and climatic data analysis softwares such as Seadas, Grads, CDO etc.

Working at “National Centre for Antarctic and Ocean Research” was one of the best learning experiences I could have. Most important thing I learn here is the process of doing a research from beginning to end. How one can get inspiration from a small talk or a lecture or news to carry forward an entire research work is the best thing that I had observed.

Working with my guide taught me how to keep learning new things everyday to keep you progressive in the field of research. Enthusiasm to dig more into the depth while analyzing a most common phenomenal to derive interesting facts about it and understand a theory related to it is the best thing I got to learn from him. He kept me motivated to work hard and learn more throughout the internship.

6.2 Cultural experience:

The best cultural experience I could ever get in Goa is to see "Goa Carnival-2018". As I was staying with a Konkani family during these 6 months, I came to know about all their food and traditions.

